INCH-POUND

MS51389B
18 January 2011
SUPERSEDING
MS51389A
20 July 1964

DETAIL SPECIFICATION SHEET

RAINCAP, FOR AIR CLEANER INTAKE, FOR INDUSTRIAL ENGINES

Inactive for new design after 22 January 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.

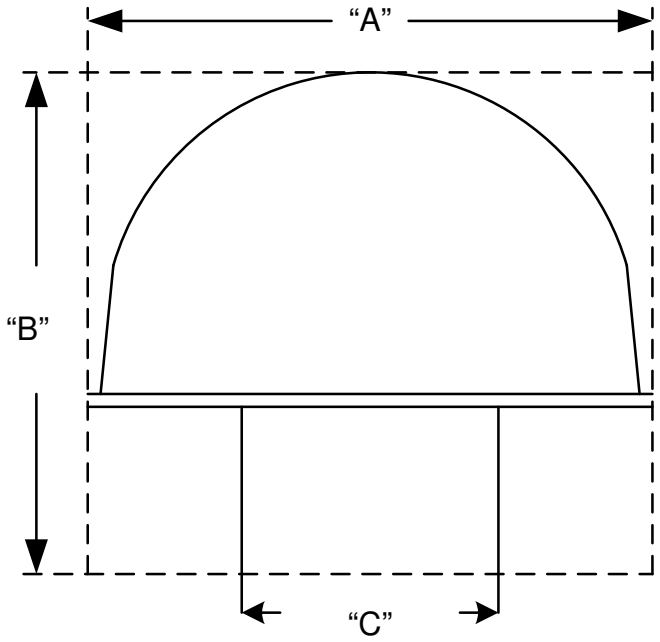

FIGURE 1. Inlet hood.

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Part or Identifying Number (PIN)	Air flow (CFM)	Maximum envelope dimension Inches (mm)		Fits outside diameter Inches (mm) ±.031 (.08)
		A	B	C
MS51389 - 1	40	5.25 (13.4)	3.43 (8.7)	1.75 (44.5)
MS51389 - 2	60	5.25 (13.4)	3.43 (8.7)	2 (50.8)
MS51389 - 3	90	5.25 (13.4)	3.43 (8.7)	2 (50.8)
MS51389 - 4	135	5.75 (14.6)	5.25 (13.4)	2.5 (63.5)
MS51389 - 5	200	7.50 (19.1)	5.93 (15.1)	3 (76.2)
MS51389 - 6	300	8.50 (21.6)	7.75 (19.7)	3.75 (95.3)
MS51389 - 7	450	9.56 (24.3)	5.06 (12.9)	4.5 (114)
MS51389 - 8	615	9.56 (24.3)	5.00 (12.7)	5 (127)
MS51389 - 9	750	9.56 (24.3)	5.00 (12.7)	4 (102)
MS51389 -10	1000	14.75 (37.5)	5.13 (13)	6 (152)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Inlet hood - Continued.

REQUIREMENTS

The design, construction, and physical dimensions shall be as specified on figure 1.

The maximum restriction when installed on an applicable air cleaner is 1.5 inches (38.1mm) of water.

The screen slot size shall be a maximum of 0.0275 square inches. (.6985 square mm)

A tapered or clamp type center tube shall be used.

Raincaps shall withstand the vibration test, specified herein, without damage that would decrease efficiency of operation.

The rain cap shall be mounted as intended in operation and subjected to method 201A vibration test of MIL-STD-202, except that the frequency shall be varied between 10 and 40 cycles per second. Vibration shall be applied horizontally in the direction of the longitudinal centerline of the air outlet. The rain cap shall be vibrated for 24 hours and any structural damage, permanent deformation or assembly breakdown during this time interval shall be cause for rejection.

Raincaps shall withstand a vacuum of 10 inches (254mm) of Hg without collapsing or other damage that would adversely affect operation.

This part is for use with air cleaners.

Referenced documents shall be of the issue in effect on the date of the invitation for bid.

For design feature purposes, this standard takes precedence over procurement documents referenced herein.

Changes from previous issue. Marginal notations are not used in this revision to identify changes due to the extent of the changes.

Referenced documents. This document references MIL-STD-202.

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 2940-2010-002)

Review activities:
Navy - MC, SA
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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